

Syllabus

Suresh L. Paul

BUS623: DATA VISUALIZATION¹

DATA VISUALIZATION USING R

INSTRUCTOR INFORMATION

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COURSE INFORMATION

Meeting room: Bloomberg Research Center Lab

Time: 6:15PM - 9:00PM on Mondays

Office hours: by appointment only

COURSE DESCRIPTION

This course introduces fundamental concepts and practical skills for data design, creation, and visualization using the R programming language. Students will explore data types and structures while learning how these foundational elements inform the principles of effective graphical data representation. Through hands-on practice, students will develop proficiency in transforming raw data into compelling and informative visualizations.

LEARNING OUTCOMES

By the end of this course, students will be able to:

- Understand and apply core data types and structures in R
- Build visualizations with the *grammar of graphics* using *ggplot2*
- Choose the right chart for a question and recognize misleading or deceptive charts
- Design accessible, interactive dashboards to communicate data to decision-makers
- Tell a clear, decision-oriented story with data
- Use AI tools to accelerate visualization work while critically supervising and correcting their output

¹This PDF document was produced from markdown using RStudio v2025.4.4.2.

COURSE MATERIALS

- *Textbook*: no assigned textbook; sample code and assignments provided by the instructor via *Canvas*
- *Datasets*: provided by the instructor
- *AI assistant*: any free tier code-producing AI assistant (e.g., *Claude*, *ChatGPT*, or *Gemini*)
- *AI coding tools (final project)*: agentic tools such as *Claude Code*, *OpenAI Codex*, or *Gemini* are introduced for building the dashboard

COMPUTING NEEDS

The Bloomberg Lab will be open during regular business hours and is equipped with all the necessary software for the course. If you plan to attend any class online, it is your responsibility to have a computer with all required software installed. If you need assistance installing R, please contact the BW IT department.

GRADING

Grade Categories	%
Homework Assignments	50
Project Presentations	30
Class Assignments & Participation	20

HOMEWORK ASSIGNMENTS

Each homework provides a dataset and a set of business questions. Homework is graded on *judgment* — chart appropriateness, accessibility, labeling, insight, and quality of AI supervision — not on whether code merely runs. Therefore, for every question you will produce:

- (a) an appropriate visualization,
- (b) a one-line justification of your chart choice,
- (c) a one-sentence insight for a decision-maker, and
- (d) an *AI Log* documenting at least one place where you corrected the AI's output.

FINAL PROJECT

Working in teams of two or three, you will build a client dashboard in R (*Shiny*) that answers a provided *client-requirements brief*. AI tools are encouraged for building it. The dashboard must run from your submitted files on a clean machine. Every team member must be able to explain and defend any part of the project during the presentation. The Final Project is assessed through the *Project Presentations* component.

USE OF AI IN THIS COURSE

AI tools such as Claude, ChatGPT, and Gemini can write visualization code in seconds. This course therefore teaches you to *direct and judge* visualization — choosing the right chart, catching a misleading one, and telling the story — skills that AI cannot supply for you. AI use is **staged**, and every assignment is labeled with its stage:

- **AI-OFF (Foundations)** — early in-class skill checks, completed on lab machines without AI or internet. Using AI here is an integrity violation, because the point is to demonstrate *your* skill.

- **AI-ASSISTED (*Disclose*)** — you may use AI, but you must log it, and you are graded on what you corrected and verified.
- **AI-ENCOURAGED (*Supervise*)** — projects and the final. Use AI freely, as professionals do; you are graded on judgment, accessibility, narrative, and how well you supervised the tool — and you must be able to defend your work live.

On *AI-assisted* and *AI-encouraged* work you must include a short **AI Log**: the tool used, what you asked it to do, what it got wrong or you changed, and what you verified yourself.

Golden rule: You may use AI to do the work faster, never to avoid understanding it. If you cannot explain and defend your work, it is not yours.

ATTENDANCE POLICY

Attendance is mandatory and essential for success in this course. You can either attend the class in-person or online via Zoom. If you are unable to attend the class live during class timings, you *may* listen to the class recordings via *Panopto* in *Canvas*. Each class content is tailored for you to successfully finish the project, which carries substantial portion of your grade. In addition, class participation carries crucial grade points. You should make efforts to attend every class during class timings unless you have a medically certified illness, death in the family, or University-sponsored activity. You are responsible for keeping up with all the lecture material and announcements made in classes.

COURSE SCHEDULE

Class 1: Intro, Concepts, Data Types & Structures, Visual Literacy
Class 2: Grammar of Graphics; base *R* plots (as contrast); intro to *ggplot2*
Class 3: *tidyverse* & *dplyr*; core plot types with *ggplot2*
Class 4: Choosing the right chart; anti-patterns; chart-lie detection; accessibility
Class 5: Interactivity & Dashboards (*Shiny*, *plotly*)
Class 6: Storytelling & Conceptualization
Class 7: Project self-work (no formal instruction; instructor available for consultation)
Class 8: Final Project presentations & live defense